EXERCISE 2: AGGREGATE FUNCTIONS & GROUPING

Database: employees_db

id	first_name	last_name	department	salary	hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

1. SELECT COUNT(id) AS Total_Number_Of_Employees
FROM employees;

Total_Number_Of_Employees
10

2. SUM() Function:

```
SELECT sum(salary) AS Total_Salary,
       department
FROM employees
WHERE department = 'IT';
```

Total_Salary	department
220000	IT

3. SELECT AVG(salary) AS Average_Salary,
department
FROM employees
WHERE department = 'HR';

Average_Salary	department
49000	HR

4. SELECT MIN(salary) AS Minimum_Salary
MAX(salary) AS Maximum_Salary
FROM employees;

Minimum_Salary	Maximum_Salary
48 000	62 000

5. SELECT SUM(salary) AS Total_Salary_Paid,
department
FROM employees
GROUP BY department;

department	Total_Salary_Paid
IT	220000
HR	99 000
Finance	119 000
Marketing	105000

6. SELECT COUNT(id) AS Total-Employees,
city
FROM employees
GROUP BY city;

City	Total-Employees
New York	2
Chicago	3
Los Angeles	2
San Francisco	2
Houston	1

7. SELECT department,
AVG(salary) AS Average_Salary
FROM employees
GROUP BY department
ORDER BY Average_Salary DESC;

department	Average-Salary
Finance	59 000
IT	55 000
Marketing	52 000
HR	49 000

8.
```
SELECT department,
       SUM(salary) AS Total_Salary_Paid
FROM employees
GROUP BY department
HAVING Total_Salary_Paid > 100 000;
```

department	Total_Salary_Paid
IT	220 000
Finance	119 000
Marketing	105 000

9.
```
SELECT city,
       COUNT(id) AS Number_Of_Employees
FROM employees
GROUP BY city
HAVING Number_Of_Employees > 1
ORDER BY city DESC;
```

City	Number_Of_Employees
Chicago	3
New York	2
Los Angeles	2
San Francisco	2

10.
```
SELECT department,
       MAX(AVG(salary)) AS Highest_Average_Salary
FROM employees;
```

department	Highest_Average_Salary
Finance	59 000